Do multi-listing hosts drag ratings down? Evidence from London Airbnb listings

Group name	[group name]
Dataset	insideairbnb.com/get-the-data: London, snapshot of 17 September 2026, file `listings.csv.gz` (CC BY 4.0)
Code	github.com/Nishant7p/DataSci-Project (notebook `notebooks/midterm_analysis.ipynb`, source in `src/`, README)

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1. Dataset

Source: Inside Airbnb (insideairbnb.com), London snapshot compiled on 17 September 2026, detailed listings file `listings.csv.gz`, downloaded on 26 October 2026 from the original site rather than from one of the older Kaggle re-uploads; the notebook checks its SHA-256. Licence CC BY 4.0. One row is one Airbnb listing.

What is raw about it: prices are text (`"$267.50"`) although Airbnb's own quotes give the currency as GBP; bathrooms exist only as text in 47 forms (`"1.5 shared baths"`); amenities are JSON lists with 9,477 distinct strings; 15 columns are completely empty; 34.9% of prices and 22.5% of ratings are missing; 179 units are listed twice; 47,950 descriptions carry HTML; 7 rows have an impossible average below 1 star; minimum stays reach 1,125 nights.

	Raw file	After encoding
Instances (rows)	91,954	91,775
Attributes (columns)	90	219
Instances × attributes	8,275,860	20,098,725

Size check: $91{,}775 \times 219 = 20{,}098{,}725 \geq 3{,}000{,}000$, so the requirement is met; the raw file already passes ($91{,}954 \times 90 = 8{,}275{,}860$). Encoding adds 33 borough, 4 room-type and 22 property-type indicators and 100 amenity indicators to 48 numeric and 12 flag columns.

Key features

Feature	Type	Real-world meaning
`review_scores_rating`	continuous	average overall guest rating, 1–5 stars; response for P1
`review_scores_accuracy`	continuous	average "did the place match its page?" score; response for P2
`host_london_listings`	discrete	listings the host runs in London (1–501), recounted after removing duplicates
`neighbourhood_cleansed`	nominal	London borough, 33 levels
`room_type`	nominal	entire home, private room, hotel room, shared room
`bedrooms`	discrete	bedrooms, 0 (studio) to 30
`price`	continuous	nightly price in GBP, £10 to £75,000
`number_of_reviews`	discrete	reviews behind the averages, 0 to 1,959
`hype_count`	discrete	marketing superlatives in the description, 0 to 10 (derived)

The full data dictionary (type, unit, range and missingness of all 90 raw and 11 derived attributes) is `docs/data_dictionary.md` in the repository.

2. Data preparation

Issue found	What we did	Why, and not the alternative
15 columns 100% empty (e.g. `license`)	dropped	Airbnb no longer exposes them; nothing to impute from
7 redundant columns (5 price quotes, 2 constant)	dropped	the quoted price equals `price` for 99.8% of listings
`price` is text, `"$267.50"`	parsed to a number in GBP	a format fix; the $ is an export artefact, the quotes say GBP
32,049 prices missing (34.9%)	kept, flagged, not imputed	92.7% were carried over from an earlier scrape and 87.3% have zero availability (0.5% of priced ones): they cannot be booked, so an imputed price would invent a market. No test uses price
20,694 ratings missing	kept, flagged; tests use only listings with 10+ reviews	no reviews means no rating, not a bad one; 10 is also the stay count Airbnb requires for Superhost
7 rows with a score average below 1	scores set to missing, flagged	a 1–5 star average cannot be below 1: these are recording errors
179 units listed twice (173 same host, name and spot; 6 same text and spot under two accounts)	extra copies dropped, the one with most reviews kept	95 copies belong to hosts with 10+ listings, the group P1 studies; left in, they would count those hosts twice
bathrooms only as text, 47 forms	parsed to a count plus shared and private flags	one string carries two facts

540 prices outside the log-price IQR fences, £19 and £1,366	capped at the fences, flagged	prices are right-skewed (median £175, max £75,000), so fences go on log price; raw-scale fences would cut ordinary luxury homes
24 minimum stays above 365 nights (max 1,125)	capped at 365, flagged	1,125 is Airbnb's form maximum, a placeholder for "long lets only"
HTML in 47,950 descriptions	tags stripped	markup is not words; needed before counting hype words
89 property types; 9,477 amenity strings	21 types with 30+ exploration rows kept, rest `Other`; 100 commonest amenities as 0/1	rare levels give unstable estimates; fences and levels are fitted on the exploration split only, so the confirmation split stays unseen

3. What changed after preparation

Measure	Before	After
Rows	91,954	91,775
Attributes	90	219
Instances × attributes	8,275,860	20,098,725
Empty columns	15	0
Duplicate units	179	0
Prices stored as text	59,905	0
Prices outside the fences	542	0 (capped)
Minimum stays above 365 nights	24	0 (capped)
Score averages below 1	7	0 (set missing)
Missing price	32,049	32,012 (kept, flagged)
Missing rating	20,694	20,628 (kept, flagged)

We lost 179 rows (duplicate units) and 22 columns (15 empty, 7 redundant), and imputed nothing. We gained a numeric price and bathroom count, a flag for every structural gap and 219 model-ready attributes. The 32,012 unpriced and 20,628 unrated listings stay in the data for the modelling stage but never enter a test.

4. Basic questions and problem statement

Split before exploring: we shuffled the 51,348 hosts with seed 558 and put 30% of them, with all their listings, in an exploration split (15,404 hosts, 27,586 listings); the other 35,944 hosts (64,189 listings) form the confirmation split. Splitting by host keeps one operator's listings on one side. Figures 1–4 use the exploration split only; Section 5 uses the confirmation split only. Every listing's side is saved in `outputs/tables/split_assignment.csv.gz`.

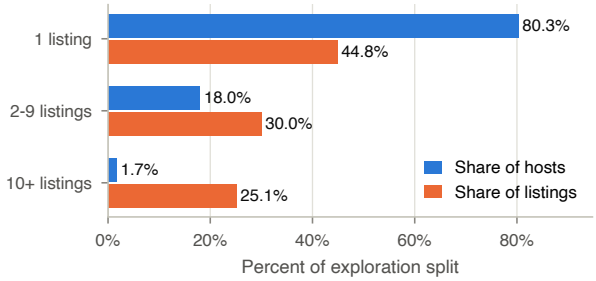


Figure 1: Hosts with 10+ London listings are 1.7% of hosts but run 25.1% of listings; four in five hosts have a single listing.

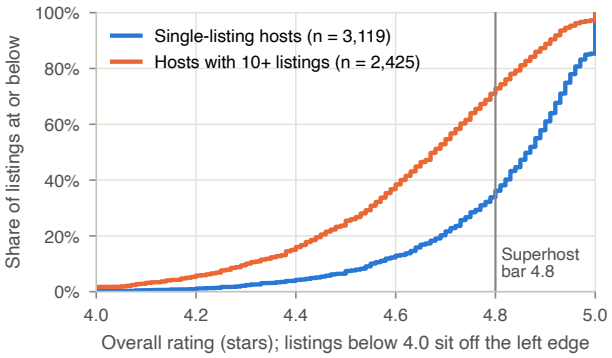


Figure 2: Among established entire homes (10+ reviews), 70.9% of large-host listings sit below the 4.8 Superhost bar, against 33.8% of single-host ones (medians 4.68 and 4.87).

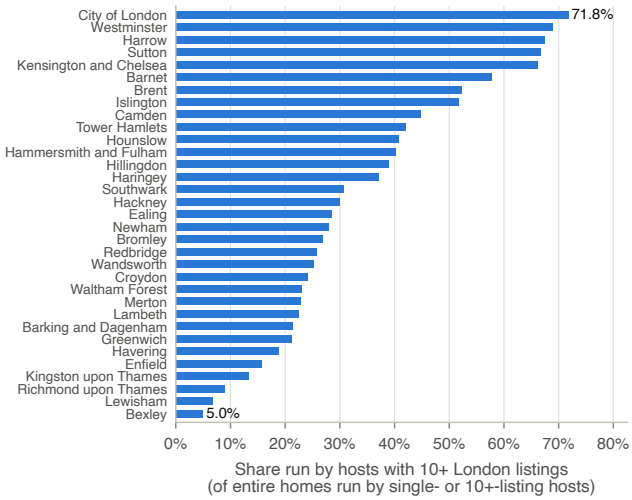


Figure 3: Of the established entire homes run by single or large hosts, large hosts run 5.0% in Bexley but 71.8% in the City of London, and mean ratings range from 4.60 to 4.87 across boroughs: borough must be held fixed.

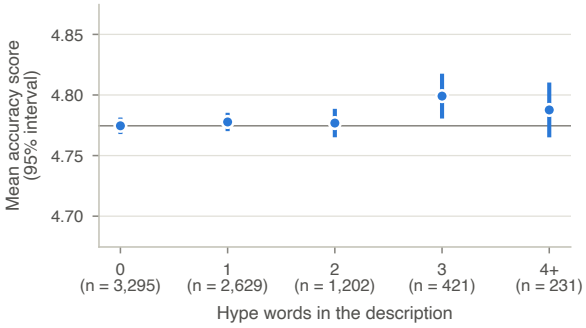


Figure 4: Accuracy hardly moves with hype words: 4.775 with none and 4.777 to 4.799 with one or more, all within 0.025 of the no-hype mean (the axis spans ±0.10 star).

Basic questions (quick answers from the exploration split)

1. Which boroughs are the most expensive? (Westminster, City of London, Kensington and Chelsea: median entire home £295–£304 a night)
2. Do entire homes cost more than private rooms? (median £224 against £80)
3. Do Superhosts get higher ratings? (4.86 against 4.68 on average)
4. Do listings with more reviews rate higher? (barely: Spearman $\rho = 0.01$)

Our problem statements

P1 (primary). Among comparable entire homes (same borough and number of bedrooms), do listings run by hosts with 10 or more London listings get different guest ratings from listings run by single-listing hosts?

P2. Among established entire homes, do listings whose descriptions pile on marketing superlatives get a different accuracy score (the guests' rating of whether the place matched its page) from plainly described ones?

How they are novel: the basic questions read one column of a listing against another. Ours depend on who runs the listing, a portfolio size that no row records directly (we recount it per host after removing duplicates), compared like for like within 99 borough-by-bedroom strata; and on a feature built from the description text, matched to the one score that measures whether a listing kept its word.

Causal flavour: P1 asks whether big operators "drag ratings down", which sounds causal. Our design only holds borough and size fixed; operators also differ in property type, pricing and guests. A causal claim would need the same property observed before and after a change of operator (quarterly snapshots as a panel) or random assignment, which a single snapshot cannot give.

5. Statistical inferences

Rows tested: the 17,929 established entire homes (10+ reviews) of the confirmation split. P1 compares the 7,197 listings of single-listing hosts (S) with the 5,288 listings of the 477 hosts with 10+ London listings (L); the 5,439 listings of hosts with 2–9 listings (a middle group, not used) and 5 without a bedroom count are left out, leaving 12,485 listings in 99 borough-by-bedroom-band strata (0–1, 2, 3+), 96 of which contain both groups. P2 keeps one random listing per host (seed 558): 2,470 *hyped* listings (2+ hype words) and 4,908 *plain* ones (none). The four tests form one family: decisions use Holm-adjusted p-values at a family-wise $\alpha = 0.05$. All settings were saved in `docs/preregistration.json` before any test ran.

Test 1 (P1): does a comparable large-host listing rate differently?

a) Hypotheses, written before running the test

H_0	$\beta = 0$
H_1	$\beta \neq 0$
Significance level	$\alpha = 0.05$ (family-wise, Holm)
Side	two-sided
Test	t-test on the within-stratum gap with host-clustered standard errors
Smallest difference that matters	0.10 star

β: gap in mean overall rating, L minus S, within borough-by-bedroom strata. Two-sided, because professional operators could plausibly rate higher (consistent service) as well as lower. 0.10 star, because the middle half of established ratings spans only 4.62 to 4.91, and 0.10 star moves a median listing (4.79) across Airbnb's 4.8 Superhost bar.

b) Other tests we considered

Two independent groups, a continuous response and an unknown σ call for a two-sample t-test, with two changes the data forces. Borough confounds host type (Figure 3), so we compare within strata (one intercept per stratum). And a large host's ratings share one operator, so they are not independent: we use host-clustered (CR1) standard errors and t with $G - 1$ degrees of freedom, G being the number of hosts.

Also considered	Why we did not use it
Welch t-test	ignores strata and clustering; gives -0.154 ($t = -39.1$, 9,512 d.f.) with a standard error a third of the clustered one (0.0039 against 0.0118)
Pooled t-test	assumes equal spreads, which Brown–Forsythe rejects (SD 0.239 against 0.182, $F = 416.9$, $p < 0.001$)
Mann–Whitney U	tests a shift in ranks, not stars, and ties at 5.0 are common; agrees in direction: $P(L > S) = 0.29$, $p < 0.001$
Stratified permutation	assumes no distribution and gives the same answer (-0.149, $p = 0.0001$, the smallest possible with 10,000 permutations), but treats listings in a stratum as exchangeable, ignoring the host clusters
z-test, paired t-test	σ unknown; nothing to pair

c) Result

Estimate	-0.145 star (raw means 4.66 for L, 4.81 for S)
95% confidence interval	-0.168 to -0.122
Test statistic (d.f.)	$t = -12.34$ (7,673)
p-value	1.1×10^{-34} (Holm: 2.3×10^{-34})
Decision	reject H_0

Comparable large-host homes rate 0.145 star lower. The whole interval lies beyond -0.10 star, so the gap matters by our standard. It also survives a conservative 476 degrees of freedom (one per large host, $p \approx 10^{-30}$).

Test 2 (P1): do large-host listings miss the Superhost bar more often?

a) Hypotheses, written before running the test

H_0	$\theta = 1$
H_1	$\theta \neq 1$
Significance level	$\alpha = 0.05$ (family-wise, Holm)
Side	two-sided
Test	Cochran–Mantel–Haenszel (CMH)
Smallest difference that matters	odds ratio outside 0.80–1.25

θ: common odds ratio, across strata, of a rating below 4.8 for L against S. 4.8 is the rating Airbnb requires for Superhost status; at the exploration single-host rate of 33.8% below it, an odds ratio of 1.25 means one more listing in twenty falls short.

b) Other tests we considered

A 0/1 outcome compared between two groups across 96 strata calls for the CMH test, which pools the strata's 2×2 tables without fitting a model.

Also considered	Why we did not use it
Pooled χ^2, two-proportion z-test	ignore the strata; give $\chi^2 = 1{,}162$ ($z = 34.1$, $z^2 = \chi^2$) and an odds ratio of 3.59, inflated by the borough mix
Fisher's exact test	meant for small counts; with 12,443 listings the χ^2 approximation is accurate
Logistic regression with stratum terms	asks the same question through a model; CMH needs none
Breslow–Day (assumption check)	rejects a common odds ratio ($\chi^2 = 223.4$, 95 d.f.), so the estimate is an average over strata; with one listing per host it does not reject ($p = 0.20$) and the odds ratio is 3.06 (2.50 to 3.73)

c) Result

Estimate	odds ratio 3.22 (65.5% of L and 34.6% of S below 4.8)
95% confidence interval	2.97 to 3.49
Test statistic (d.f.)	$\chi^2_{\mathrm{MH}} = 838.2$ (1)
p-value	2.7×10^{-184} (Holm: 1.1×10^{-183})
Decision	reject H_0

Within the same borough and bedroom band, a large-host listing has 3.2 times the odds of falling below the Superhost bar. The whole interval lies above 1.25, so the difference matters.

Test 3 (P1): with every host counted once, do large hosts rate lower?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_L - \mu_S = 0$
$\boldsymbol{H_1}$	$\mu_L - \mu_S \neq 0$
Significance level	$\alpha = 0.05$ (family-wise, Holm)
Side	two-sided
Test	Welch t-test
Smallest difference that matters	0.10 star

μ_L, μ_S: mean rating of a host's established entire homes, averaged per host (477 large hosts, 7,197 single hosts). Averaging removes the dependence within a host by design rather than by a model.

b) Other tests we considered

Two independent groups of hosts, very unequal in size and spread: Welch's t-test.

Also considered	Why we did not use it
Pooled t-test	Brown–Forsythe rejects equal variances ($F = 9.58$, $p = 0.002$); run anyway, it gives $t = -18.96$ and the same decision
Mann–Whitney U	tests ranks, not stars; agrees: $P(L > S) = 0.24$, $p < 0.001$
Permutation test	also valid; $p = 0.0002$, the smallest two-sided value 10,000 resamples can give
Stratified tests	a host's listings span several boroughs, so host means cannot be stratified; Tests 1 and 2 cover that

c) Result

Estimate	-0.164 star (host means 4.65 for L, 4.81 for S)
95% confidence interval	-0.182 to -0.146
Test statistic (d.f.)	$t = -17.85$ (532.4)
p-value	3.4×10^{-56} (Holm: 1.0×10^{-55})
Decision	reject H_0

The typical large operator rates 0.16 star below the typical single host. The interval lies beyond -0.10 star, so this gap matters too.

Test 4 (P2): do hyped descriptions change the accuracy score?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\delta = 0$
$\boldsymbol{H_1}$	$\delta \neq 0$
Significance level	$\alpha = 0.05$ (family-wise, Holm)
Side	two-sided
Test	stratified permutation test (10,000 permutations)
Smallest difference that matters	0.10 star

δ: gap in mean accuracy score, hyped minus plain, averaged over borough-by-host-class strata. Hyped means 2 or more words from a fixed list of 34 superlatives ("perfect", "stunning", "luxury", ...): the largest count still covering 15% of exploration listings (23.8% have 2+). Two-sided, because the exploration split showed no direction (Figure 4). 0.10 star, because Airbnb shows category scores such as accuracy to one decimal.

b) Other tests we considered

Accuracy averages are bounded and bunched near 5 (median 4.83), so a permutation test avoids assuming normality; permuting labels within strata holds borough and operator type fixed; and one listing per host stops big operators' copied descriptions from counting many times.

Also considered	Why we did not use it
Welch t-test	ignores the strata; $+0.005$, $p = 0.28$
Mann–Whitney U	ranks with many ties, and ignores the strata; $p = 0.25$
Unstratified permutation	same as ours without the strata; $p = 0.28$
Spearman correlation	uses every hype count (a dose–response check), but no strata; $\rho = 0.017$, $p = 0.08$ ($n = 10{,}513$)

c) Result

Estimate	$+0.008$ star (4.825 hyped, 4.821 plain)
95% confidence interval	$+0.0004$ to $+0.016$ (stratified bootstrap, 5,000 resamples)
Test statistic (d.f.)	the stratified gap itself; d.f. not applicable (permutation test)
p-value	0.044 (Monte Carlo SE 0.002; Holm: 0.044)
Decision	reject H_0

Hyped listings score very slightly higher on accuracy, the opposite of the "overselling backfires" idea, and only once borough and host class are held fixed: the unstratified tests

give $p \approx 0.28$. The whole interval lies inside ± 0.10 star, so whatever the gap is, it is far too small to matter.

Summary of the four tests

Test	Estimate (95% CI)	Holm p	Verdict
T1	-0.145 star (-0.168, -0.122)	2.3×10^{-34}	reject; matters
T2	odds ratio 3.22 (2.97, 3.49)	1.1×10^{-183}	reject; matters
T3	-0.164 star (-0.182, -0.146)	1.0×10^{-55}	reject; matters
T4	$+0.008$ star ($+0.0004$, $+0.016$)	0.044	reject; too small to matter

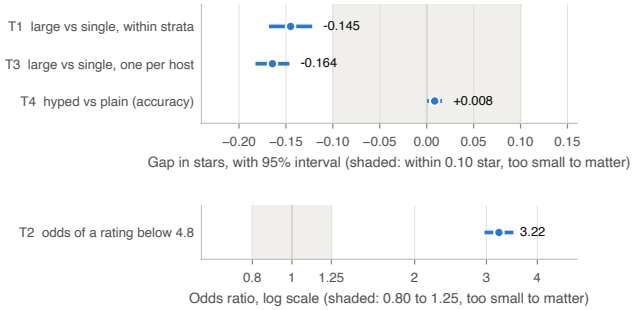


Figure 5: The three P1 estimates lie well outside their "too small to matter" bands (shaded), while the P2 gap sits inside its band.

For P1, three designs that handle the data's two problems in different ways (strata for the borough mix, clusters or host-level units for repeated operators) agree: a large-host home rates about 0.15 star lower than a comparable single-host home and has about three times the odds of missing the Superhost bar. For P2, hype in a description makes no difference a guest could see.

6. Limitations

- **Association, not cause.** Large operators differ from single hosts in more than size (property type, pricing, guests, house rules); our strata hold only borough and bedrooms fixed.
- **Established entire homes, one snapshot.** The tests cover entire homes with 10+ reviews on 17 September 2026, not private rooms, new listings or listings that had already left Airbnb.
- **Compressed ratings.** Half of established ratings lie between 4.62 and 4.91, and each average rests on a different number of reviews; our tests do not weight by that precision.
- **Dependence handled only in part.** Test 1 clusters by host, but only 477 clusters carry the treatment, so its 7,673 d.f. may be generous (with 476 d.f. the result stands). Test 2 treats listings as independent; with one listing per host its odds ratio is 3.06.
- **A pooled odds ratio over unequal strata.** Breslow–Day rejects a common odds ratio, so 3.22 averages different stratum effects.
- **Missing prices are not random.** 35% of listings have no price because they cannot be booked; conclusions about price would not generalise, which is why no test uses it.
- **A crude hype measure.** A fixed English word list misses context ("perfect for families") and other languages, and host size counts London listings only (Airbnb-wide portfolios reach 5,777).